

CREM collapse survival and experimental comparison

Survival fraction

1
0.8
0.6
0.4
0.2
0

10^{-3}

10^{-2}
 t [ns]

CREM trajectories: $N = 100$; valid = 100

Full CREM calibration; secular $P(a) \propto a^{-4}$

cutoff: $r = 0.01 a_0$; all orbits start at $r = a_0$

$\langle t_{\text{collapse}} \rangle = 0.0030203 \pm 0.0001385$ ns (SE of mean)

sample $\sigma/\langle t \rangle = 0.45856$ (narrow, NOT exponential)

blue dashed: descriptive $\exp()$ through t - shape

is illustrative only, the sample is not exponential.

External comparison only:

$\tau_{\text{exp}} = 142.04 \pm 0.025836$ ns ($\tau_{\text{exp}} / \langle t \rangle \approx 4.7 \times 10^4$)

p-Ps and o-Ps share this classical inspiral: they differ

only by initial dipole alignment ($\epsilon 10^{-5}$ of Coulomb),

while τ_{exp} differs by $\approx 10^3$. Scale reference,

not a prediction of the annihilation lifetime.

Vallery et al., PRL 90 (2003)